



MONASH University

Assessment 1

Faculty of Information Technology

FIT5125 / FIT4005

IT Research and Innovation Methods

Semester 2, 2025

Assignment 2

Topic

Tasks related to material in Weeks 7-12.

Value

This assignment is worth **45%** of the total marks for FIT5125/FIT4005.

Assignment due date

11:55 PM Friday 7th November 2025 (AET)

Submission method

Submit the following in Moodle:

- Three PDF documents, one for each of the three tasks (A, B and C).

Assignment Criteria:

Task A Descriptive Statistics	Task B Inferential Statistics	Task C Design
Weighting: 30%	Weighting: 30%	Weighting: 40%
See Page 3.	See Page 4.	See Page 5.

This is an individual assessment; it must be your own work and expressed in your own words.

☞ *A marking rubric is available on Moodle.*

☞ *There are specific requirements for file names on your submission.*

Assessment rules:

1. Note that plagiarism detection procedures may be applied to each submission. See the University rules and regulations regarding plagiarism and resulting penalties. Any case of plagiarism detected will result in the automatic failure of the entire assignment.
2. Late submissions will incur a penalty of 5% per day, see:
<https://publicpolicydms.monash.edu/Monash/documents/1935752>
3. Monash policy on Special Consideration is available at:
<https://www.monash.edu/exams/changes/special-consideration>
4. Due to the size of the unit (over 1000 students) and the nature of the assignment, we aim to mark and return work within **15 working days** of submission (i.e. **21 November 2025**).
5. AI tools (e.g., ChatGPT) **are permitted** for this assignment, and there is **no requirement** to include a statement regarding their use.

IMPORTANT

- Any **questions** about the assignment should be submitted as a public post to the **Ed forum** (under the sub-category “Assignments”) so that all students have access to your question and the Chief Examiner’s or Unit Coordinator’s response.
- Task A and B are **individual activities**, and elements of Task C may be completed as either an **individual** or a **group** (2-4 students); however, you must follow Monash University’s policies, procedures and regulations relating to academic integrity, plagiarism and collusion.
- This is a **formal assessment**, so tutors are not permitted to provide direct support to you. However, they can give feedback on related studio activities (during a studio).
- This assignment aims to evaluate your understanding of descriptive and inferential statistics in research, and the application of human-centred design methods..

Task A (Week 8)

“Descriptive Statistics: Telling a Data Story”

Understanding patterns in **system usage** is essential for **designing better systems**. In this task, you will **analyse real data from the Kluster Networking Challenge**, which investigated **how different system configurations affect participant engagement and interaction patterns**.

The Kluster study tested **four experimental conditions**:

- All Features + 50% Waiting List
- All Features + No Waiting List
- Without YouTube + 50% Waiting List
- Without YouTube + No Waiting List

Your goal is to understand the data from the perspective of designing a better collaborative system.

Complete the following:

1. **Explore the Kluster Networking Challenge datasets** provided on Moodle (system logs and transcription logs). Familiarise yourself with the available variables and data structure (no submission).
2. **Formulate a research question** that examines differences between the four experimental conditions. Your question should focus on **participant engagement** and **interaction patterns** (e.g., how features and waiting lists impact behaviours like "hopping around" between rooms, session duration, feature usage, etc.) **(max. 75 words)**.
3. Select and calculate **two appropriate descriptive metrics** (e.g., mean, median, standard deviation, frequency counts) that reveal differences across ALL four conditions. Present:
 - The variables/fields you used from the dataset
 - The calculated values for each condition
 - A brief justification of why these metrics are appropriate **(max. 100 words)**.
4. Create and submit an **appropriate visualisation** that clearly shows the differences between all four experimental conditions. The visualisation should:
 - Be fully annotated (title, axis labels, legend, units)
 - Use the appropriate chart type for your data
 - Make comparisons between conditions easy to interpret
5. Write a **narrative description** of your findings as they relate to your research question. Your narrative should:
 - Reference both your chosen metrics and visualisation
 - Discuss what the differences (or similarities) between conditions suggest about participant engagement and interaction patterns
 - Connect findings to implications for designing better systems **(max. 250 words)**.

What to Submit

A PDF document, named "**YOUR-STUDENT-ID-Assignment-2-Task-A.pdf**", containing your response to the assignment (including the visualisation)..

How Much to Write

Follow the maximum word limit stated above. The word count does not include spaces. Only words within the word limit will be marked.

Task B (Week 9)

“Inferential Statistics: Working with Hypotheses”

Communication patterns in collaborative systems can vary based on numerous factors, including system features, participant characteristics, and interaction dynamics. In this task, you will **formulate and test hypotheses about communication patterns** using a subset of the **Kluster Networking Challenge communication data**.

You will be provided with a subset of data, including variables related to:

- participant communication style
- relevant communication metrics
- participant demographics

Use **inferential statistics** to **test relationships or differences in communication patterns**:

1. Explore the **Kluster Networking Challenge communication and survey data subset** provided on Moodle. Familiarise yourself with the available variables (no submission).
2. Formulate and write down a **testable hypothesis** about participants' **communication behaviours** by linking variables from the **Kluster Networking Challenge communication data subset** with measures from the **survey dataset**. The hypothesis should be specific and suitable for examination using inferential statistical methods (**max. 75 words**).
 - Focus on relationships between variables or differences between groups.
 - Be realistic that the relationship can be tested with the data provided.
 - Relate to communication styles, participation patterns, or demographic factors.
3. Write down the **null hypothesis** for your proposed hypothesis (**max. 50 words**).
4. **Identify your variables (max. 75 words)**:
 - Independent variable(s).
 - Dependent variable(s).
 - At least two confounding variables that could affect your results.
5. **Describe your statistical approach (max. 200 words)**:
 - What statistical test(s) will you use to test your hypothesis?
 - What are your assumptions about the data (e.g., normality, scale of measurement)?
 - Why is/are this/these test/tests appropriate for your hypothesis and data?
6. Conduct the statistical test(s) using the provided data and a tool of your choice (e.g., Python, R, SPSS, Excel). Present (**max. 150 words**):
 - The test statistic(s) and p-value(s).
 - Your interpretation of the results (do you reject or fail to reject the null hypothesis?)
 - What this means in practical terms for understanding communication in Kluster.

What to Submit

A PDF document, named "**YOUR-STUDENT-ID-Assignment-2-Task-B.pdf**", containing your response to the assignment.

How Much to Write

Follow the maximum word limit stated above. The word count does not include spaces. Only words within the word limit will be marked.

Task C: Weeks 11 and 12 (Design)

“Design for Multilingual Collaboration”

Design is a critical component of the innovation process. In this task, you will **develop design problem statements** based on the **bank of user quotes** provided on Moodle. These quotes capture real experiences of people using Zoom, focusing on **challenges and opportunities for multilingual interaction and online engagement in Zoom**.

Using insights from the quotes, you must **identify and describe key interaction problems** experienced by users. Based on this analysis, you will **propose potential design solutions** that address these problems and improve users’ overall interaction experience in Zoom. Your ideas should make online interaction more **inclusive, natural, and socially aware** for participants from diverse linguistic backgrounds.

For this activity, you may work alone or in a group of 2-4 students (e.g. from Kluster Networking Challenge) when brainstorming and affinity diagramming; however, the design problem statements, design rationales, and the annotations to the Miro boards that you submit must be your own work.

Step 1: Understanding the Design Context

1. Review the bank of quotes provided on Moodle. These quotes are accounts of user experiences with video collaboration platforms, with particular focus on multilingual contexts.
2. Using **Miro**, develop an affinity diagram to identify and synthesise key themes emerging from the provided quotes that relate to multilingual collaboration. Group similar ideas, patterns, and user sentiments together to uncover the main challenges, needs, frustrations, and opportunities expressed by users in their experiences with online interaction.
3. Based on the themes you identify, develop two or more design problem statements that you will use to brainstorm solutions.

Step 2: Design Ideation

4. Using **Miro**, brainstorm solutions to address your problem statements, consider:
 - ◆ What extensions to the features or interface elements on Zoom would help?
 - ◆ How would these features or interface elements work?
 - ◆ How do they address the needs expressed in the user quotes?
5. Identify two **significantly different** solutions; extend the description of these designs on your Miro board and separately write a design rationale statement for each to capture:
 - ◆ How the design addresses the needs and challenges identified in the user quotes.
 - ◆ What aspects does your design aim to support or improve?
 - ◆ Potential limitations or trade-offs in your design.

What to Submit

1. Individual submission:
 - A PDF document named "**YOUR-STUDENT-ID-Assignment-2-Task-C.pdf**"
2. Your PDF should contain:
 - Links to annotated copies of your Miro boards for Part 1 and Part 2; annotations should allow a marker to understand your affinity diagramming and brainstorming process..
 - Your design problem statements (**max 50 words per problem statement**)
 - Your two design rationale statements (**max 400 words per rationale**)